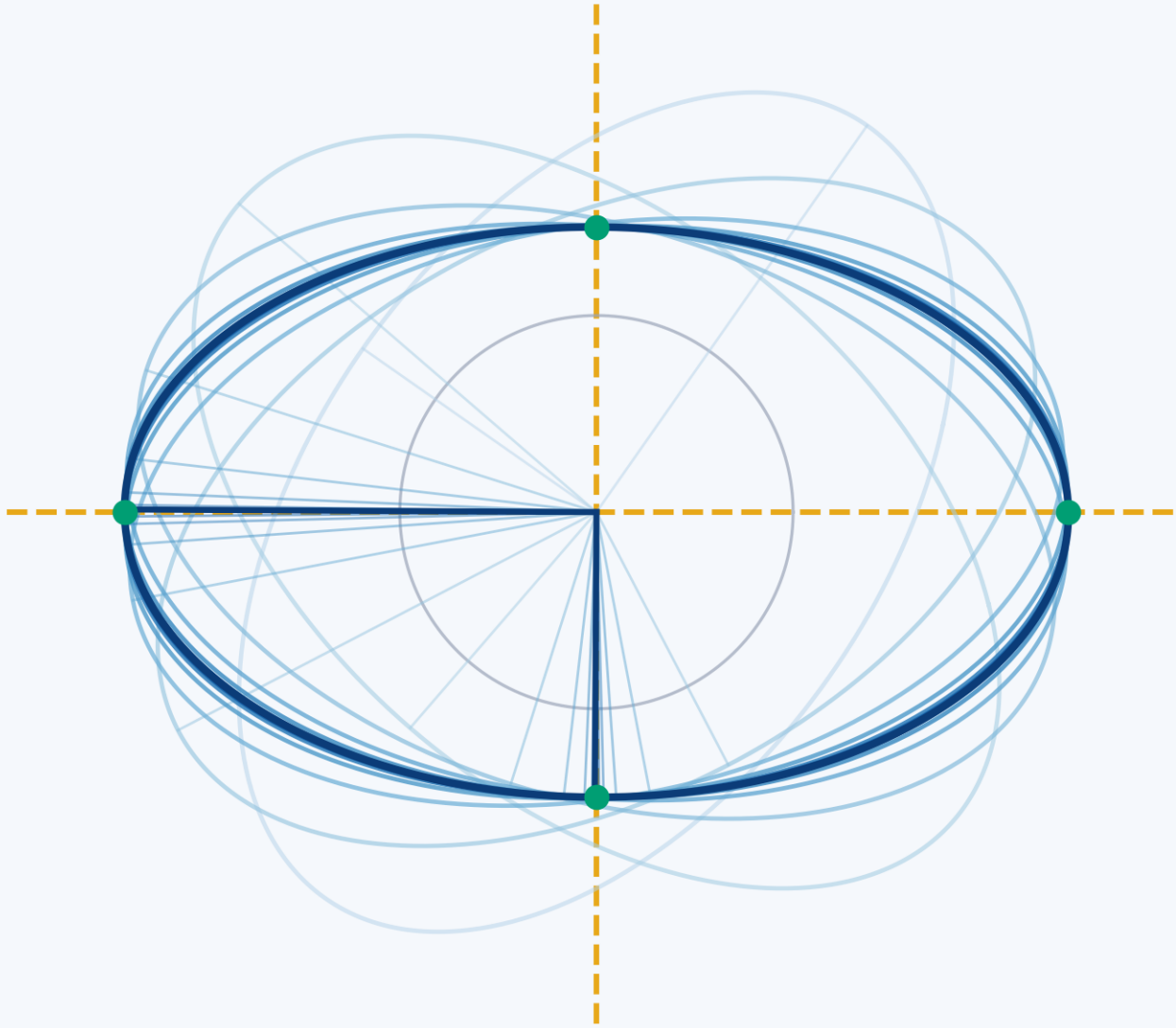
Eigenvalue Demos: QR Iteration with Shifts

Companion to the interactive module



How to use the interactive module

Eigenvalue Problems

Heath Chapter 4 · interactive modules

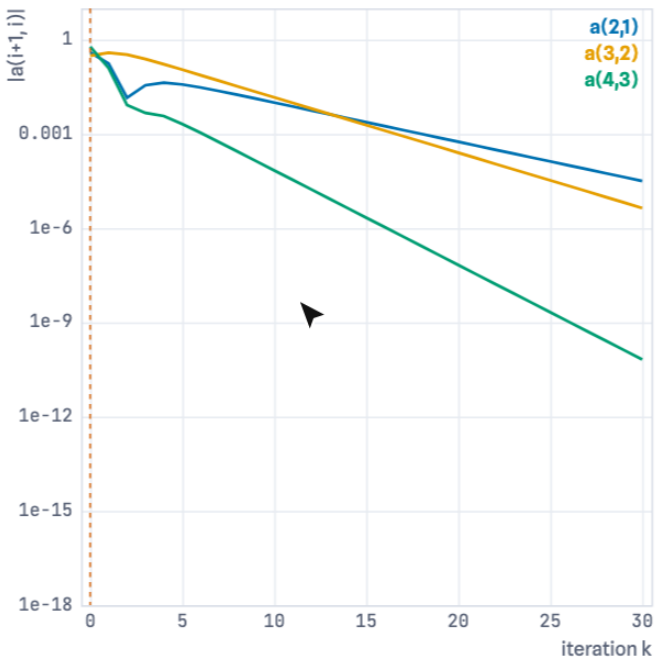
Power and inverse iteration | **QR iteration with shifts** | Hessenberg reduction | Jacobi's method

A_k (SUBDIAGONAL OUTLINED)

A_0

2.9766	0.3945	0.4198	1.1159
0.3945	2.7328	-0.3097	0.1129
0.4198	-0.3097	2.5675	0.6079
1.1159	0.1129	0.6079	1.7231

|SUBDIAGONAL ENTRIES| VERSUS ITERATION



QR iteration with shifts

Examples: choose an example...

Size n: 4 × 4

A

2.9766	0.3945	0.4198	1.1159
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Shift: no shift

☐ reduce to Hessenberg form first

◀ Back | Step ▶ | +10 | Restart

Diagonal and shift

```
k = 0   shift used for the next step: 0
diagonal of Aₖ: 2.9766, 2.7328, 2.5675, 1
```

QR iteration: factor, multiply in reverse order, repeat

$$A_k = Q_k R_k, \qquad A_{k+1} = R_k Q_k = Q_k^T A_k Q_k$$

$$A_k - \sigma_k I = Q_k R_k, \qquad A_{k+1} = R_k Q_k + \sigma_k I$$

$$|a_{i+1,\,i}^{(k)}| \sim |\lambda_{i+1}/\lambda_i|^k \quad \text{(no shift)}$$

- each step is an orthogonal similarity: same eigenvalues
- $A_k \rightarrow$ triangular (diagonal if A is symmetric), largest λ first
- Rayleigh shift $\sigma_k = (A_k)_{nn}$: fast last row, then deflate

In the demo

- Examples: the 4 × 4 matrix with λ = 4, 3, 2, 1
- Step: A_k updates, subdiagonal outlined
- plot: |subdiagonal entries| on a log scale
- Shift: none, Rayleigh or Wilkinson
- readout: diagonal of A_k vs the eigenvalues

Every step is a similarity transformation, so QR iteration only reorganizes A until the eigenvalues can be read off the diagonal.

Worked example: one QR step by hand

Example [3 1; 1 3] with no shift. Gram–Schmidt gives Q_0 and R_0; then multiply in reverse order.

Matrix

$$A_0 = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

Factor $A_0 = Q_0 R_0$

$$Q_0 = \begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix} / \sqrt{10}, \quad R_0 = \begin{bmatrix} 10 & 6 \\ 0 & 8 \end{bmatrix} / \sqrt{10}$$

Reverse product

$$A_1 = R_0 Q_0 = \begin{bmatrix} 3.6 & 0.8 \\ 0.8 & 2.4 \end{bmatrix} \quad \text{tr} = 6, \ \det = 8$$

The subdiagonal fell from 1 to 0.8; trace and determinant (so the eigenvalues 4 and 2) are unchanged. Later steps shrink it by $\lambda_2/\lambda_1 = 1/2$.

What the demo draws: A_k with the subdiagonal outlined

A_0

2.977	0.395	0.420	1.116
0.395	2.733	-0.310	0.113
0.420	**-0.310**	2.567	0.608
1.116	0.113	**0.608**	1.723

A_5

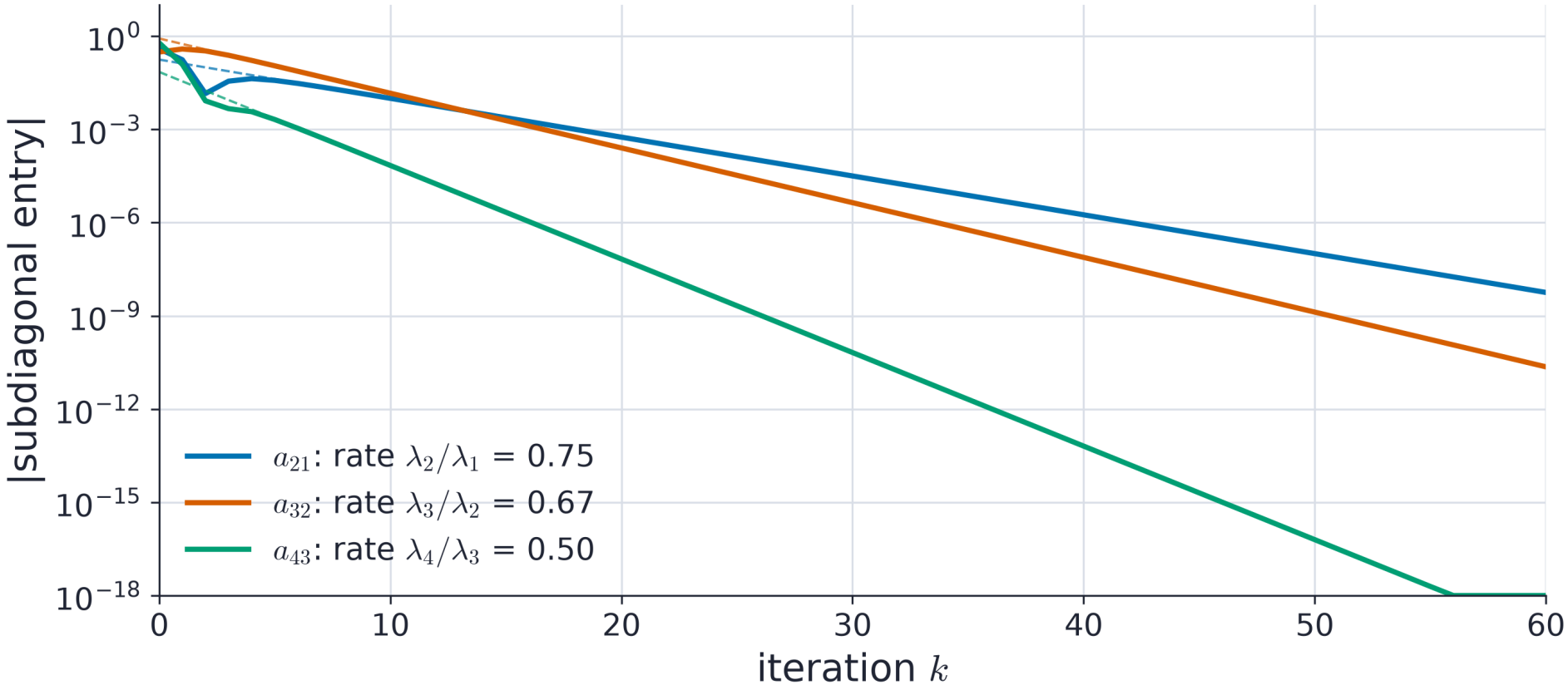
3.998	-0.038	0.042	-0.002
-0.038	2.989	-0.111	0.001
0.042	**-0.111**	2.013	0.002
-0.002	0.001	**0.002**	1.000

A_{30}

4.000	-3.2e-5	1.2e-9	0
-3.2e-5	3.000	-4.4e-6	-1.2e-15
1.2e-9	**-4.4e-6**	2.000	-6.6e-11
0	-1.1e-15	**-6.6e-11**	1.000

The 4 × 4 example, no shift: after 30 steps the diagonal reads 4, 3, 2, 1 and the subdiagonal (orange) is small. The outlined entries are the ones plotted on the next slide.

Without shifts: each subdiagonal entry decays at its own rate

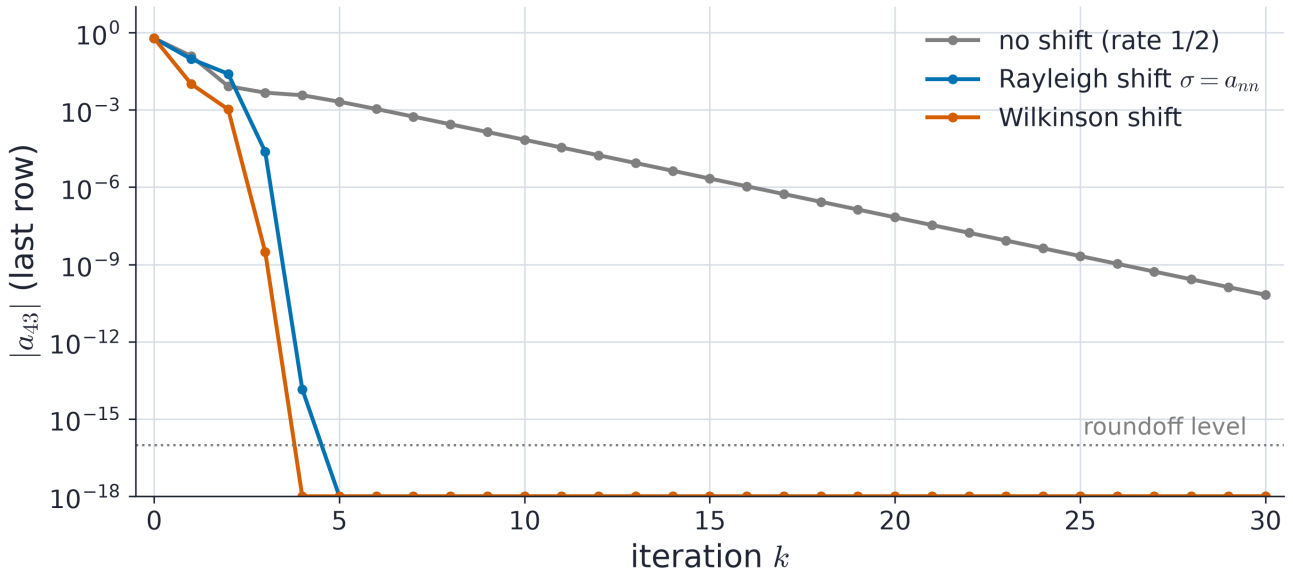


Entry (i+1, i) decays like $|\lambda_{i+1}/\lambda_i|^k$: 3/4, 2/3 and 1/2 here. Linear, and slow when neighbouring eigenvalues are close.

Shifts: make the last row converge fast, then deflate

$$\text{Rayleigh}: \quad \sigma_k = a_{nn}^{(k)}$$

$$\text{Wilkinson}: \quad \sigma_k = \text{eigenvalue of the trailing } 2 \times 2 \text{ block nearest } a_{nn}^{(k)}$$



Deflation

- $|a_{43}| < 10^{-15}$ after 5 steps (Rayleigh) or 4 (Wilkinson), versus about 50 unshifted
- a_{44} is then the eigenvalue 1
- continue on the 3 × 3 block

A shift near an eigenvalue makes the bottom entry of $A_k - \sigma I$ tiny, so one QR step nearly isolates that eigenvalue: the same idea as inverse iteration.

When the Rayleigh shift fails, and Wilkinson's fix

Matrix [3 1; 1 3] again: both diagonal entries are 3, exactly halfway between the eigenvalues 2 and 4.

Rayleigh shift

$$\sigma = a_{22} = 3: \quad A_0 - 3I = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = Q_0 \cdot I \Rightarrow A_1 = Q_0 + 3I = A_0$$

Wilkinson shift

$$\text{Wilkinson}: \ \sigma = 4 \Rightarrow A_1 = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \ (\text{up to signs})$$

σ = a_{nn} can sit exactly between two eigenvalues and QR iteration stalls ($A_1 = A_0$). Wilkinson's shift uses the trailing 2 × 2 block and avoids this.

Two more things the demo shows

A_{60} (no shift)

2.052	0.263	-0.025
0	-0.048	-1.009
0	**1.038**	-0.004

Complex eigenvalues: a 2 × 2 block stays

- $a_{21} \to 0$ and $a_{11} \to 2.052$, but a_{32} keeps oscillating near ±1
- the purple 2 × 2 block holds the pair −0.026 ± 1.023i

"Reduce to Hessenberg form first"

- zeros below the first subdiagonal; QR iteration keeps that form
- each step then costs $O(n^2)$ instead of $O(n^3)$; symmetric A becomes tridiagonal

Practical QR iteration = Hessenberg reduction + shifts + deflation. This is what numpy.linalg.eig calls (LAPACK).

What the demo shows: QR iteration on the 4 × 4 example

Eigenvalue Problems

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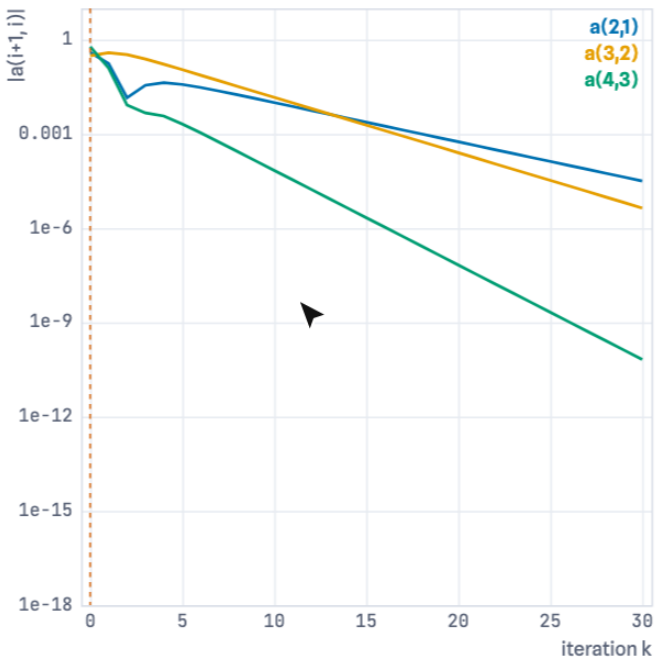
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QR iteration with shifts

Examples: choose an example...

Size n: 4 × 4

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Shift: no shift

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Diagonal and shift

```
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```

What the demo shows: A_k for the other three items

Eigenvalue Problems

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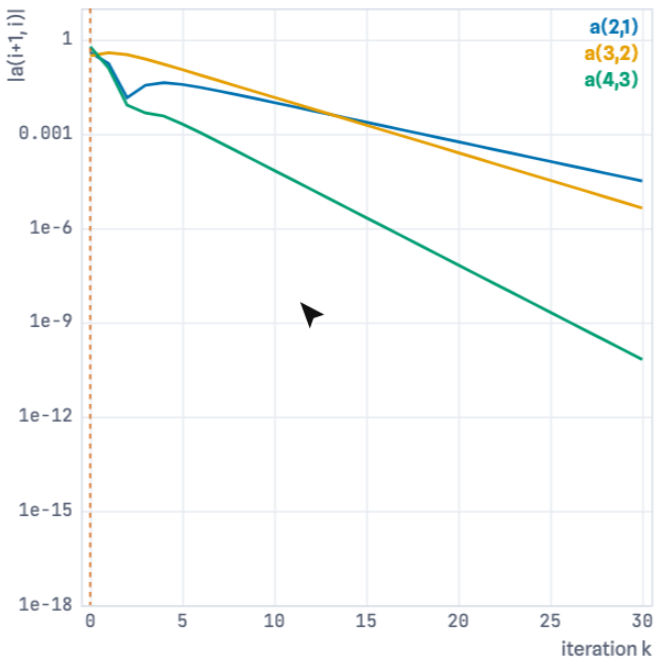
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QR iteration with shifts

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Diagonal and shift

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diagonal of Aₖ: 2.9766, 2.7328, 2.5675, 1
```

Common mistakes

No shift, close eigenvalues

- each entry decays like $|\lambda_{i+1}/\lambda_i|$
- slow when neighbouring λ are close: use a shift

The Rayleigh shift stalls

- $\sigma = a_{nn}$ halfway between two eigenvalues: $A_1 = A_0$
- the Wilkinson shift avoids this

Complex eigenvalues on the diagonal?

- a complex pair stays as a 2×2 block
- its eigenvalues are the pair

Never deflating

- $|a_{n,n-1}|$ at roundoff: a_{nn} is an eigenvalue
- drop the last row and column, continue

Summary: which method does what

method	finds	cost per step	convergence
power iteration	largest $\|\lambda\|$	one product Ax	linear, $\|\lambda_2/\lambda_1\|$
inverse iteration (σ)	λ nearest σ	two triangular solves (LU once)	linear, $\|\lambda c - \sigma\| \,/\, \|\lambda n - \sigma\|$
Rayleigh quotient iteration	a λ near the start	new LU each step	cubic (symmetric)
QR iteration, no shift	all λ	a QR factorization	linear, $\|\lambda_{i+1}/\lambda_i\|$ per entry
QR iteration with shifts	all λ, bottom row first	$O(n^2)$ on Hessenberg form	quadratic or cubic, then deflate

One idea links them: shifting toward an eigenvalue makes the matching component dominate, in a vector (inverse iteration) or in the last row of A_k (shifted QR).